

Abdullah Alfateel

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PROFESSIONAL SUMMARY

Computer Science graduate (KFUPM, Second Honor Distinction, GPA 3.48/4.0) with full-stack, systems, and AI experience. Built production-grade applications at Saudi Aramco using React and TypeScript; designed and deployed an end-to-end AI backend integrating FastAPI, RAG, and LLM orchestration; and developed real-time edge AI for IoT environments. Comfortable across software development, AI and ML engineering, and systems infrastructure. AWS Certified AI Practitioner with three Oracle Cloud credentials.

TECHNICAL SKILLS

Programming & Software: Python, JavaScript, TypeScript, React, Node.js, HTML/CSS, Tailwind, Bootstrap, SQL, R, Java, C, RESTful API design

AI / ML & Data: Scikit-learn, PyTorch, TensorFlow, HuggingFace, RAG pipeline design, ChromaDB, Ollama / Llama 3, FastAPI, Pandas, NumPy, Power BI, Tableau

Systems & Infrastructure: Linux, Git, Raspberry Pi (edge inference, IoT), computer vision, IoT data pipelines, networking (Huawei Core Switches, Aruba, iMaster NCE-Campus)

Cloud & Tools: AWS (Certified AI Practitioner), Oracle Cloud (GenAI Professional, AI Vector Search, AI Foundations), Google Colab, Enterprise Architect (UML)

EXPERIENCE

Saudi Aramco

Computer Science Intern | Dhahran, Saudi Arabia | June 2024 – August 2024

- Designed and built an internal innovation tracking dashboard (React, TypeScript) from requirements to delivery, translating project lifecycle data into structured KPIs for senior stakeholders.
- Developed front-end components (JavaScript, Bootstrap) for an internal web platform, improving cross-team information access and collaboration.
- Supported enterprise network infrastructure (Huawei Core Switches, Aruba, iMaster NCE-Campus) in a professional R&D data centre environment.
- Recognised with a formal recommendation letter for technical performance and cross-functional professionalism.

KEY PROJECTS

AI Assessment Generation System (RAG-Based) – Personal Project

- Designed and built a full-stack AI system end-to-end: FastAPI backend, automated GitHub code ingestion, semantic chunking, HuggingFace embeddings, ChromaDB vector storage, and Ollama / Llama 3 LLM orchestration with JSON schema-validated outputs.
- Architected for local self-hosted deployment with no third-party API dependencies, enforcing data quality and PII safety at every pipeline stage.

EEG-Controlled Prosthetic Arm (Capstone Project)

- Led AI development within a multidisciplinary team: designed and executed a full ML pipeline (preprocessing, feature extraction, model training, evaluation) achieving approximately 72% accuracy on a 6-class IEEE benchmark dataset using ensemble methods.
- Deployed the production model on Raspberry Pi for real-time inference controlling a 3D-printed prosthetic arm, handling edge compute constraints, latency requirements, and hardware integration.

Smart Road Safety System (AgentIQ Bootcamp – NITA)

- Developed computer vision and edge AI for pedestrian monitoring and jaywalking detection on a Sony camera and Raspberry Pi device, integrated into a live IoT pipeline serving transport authority dashboards.
- Collaborated with industry partners (Sony, PipeLogic, Cumulocity) across system design, data pipeline integration, and stakeholder reporting.

EDUCATION & CERTIFICATIONS

- B.S. Computer Science | KFUPM | May 2025 | GPA: 3.48 / 4.00, Second Honor Distinction
- AWS Certified AI Practitioner
- Oracle Cloud Infrastructure: Generative AI Professional, AI Vector Search Certified Professional, AI Foundations Associate
- Tableau Fundamentals (Coursera) | KAUST Academy Advanced AI (2024)

LANGUAGES

- Arabic – Native
- English – IELTS Academic Band 7.5